Pushkar Jain

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Professional Summary

Software Engineer specializing in scalable backend systems and AI-powered platforms. Experienced in designing distributed architectures, microservices, and real-time processing pipelines. Proven ability to take systems from ideation through benchmarking to production deployment with focus on performance, reliability, and observability.

Technical Skills

Backend & Systems: Python, FastAPI, gRPC, REST APIs, Microservices, AsyncIO, Distributed Systems
System Design: Scalability, High-throughput Systems, API Design, Caching, Load Balancing, Fault Tolerance, Observability
AI/ML & LLMs: Multi-Agent Systems, LangChain, LangGraph, Claude Sonnet, Google Gemini, Prompt Engineering
Data & Processing: Pandas, NumPy, OpenCV, Vector Search (FAISS, Chroma), Real-time Pipelines
DevOps & Tools: Docker, Prometheus, Grafana, Playwright, FFmpeg, Git, Linux
Testing & Security: Pytest, JWT Authentication, TLS, eBPF, Validation Frameworks

Experience

AI/ML Engineer Intern — Jul 2025 – Present
Docu3c — *Newcastle, Washington*

- Designed production-grade multi-agent orchestration systems using FastAPI, LangGraph, and Azure for real-time AI workflows.
- Architected scalable backend services with microservice patterns enabling modular agent coordination and independent scaling.
- Implemented async pipelines (AsyncIO, concurrent workers) reducing response latency under concurrent workloads by 35%.
- Built observability stack (Prometheus + Grafana) for real-time monitoring and performance tuning.
- Designed fault-tolerant systems with retry mechanisms, validation layers, and caching, improving cost efficiency by 40%.
- Led architecture discussions and made trade-offs between latency, cost, and scalability for production AI systems.

Projects

Video Forge – Deterministic LLM Video Synthesis Pipeline — Mar 2026
Claude Sonnet 4.5, GSAP, Playwright, FFmpeg, SerpAPI, Streamlit

- Engineered end-to-end high-throughput distributed pipeline converting natural language to 1080×1920 MP4 videos via multi-stage architecture.
- Designed deterministic processing system with validation layers achieving 100% first-pass success rate (zero retries).
- Implemented parallel processing (4 Playwright workers) ensuring byte-level consistent outputs (SSIM = 1.000).

Multi-Agent AI Orchestration System — May 2025
LangGraph, gRPC, FastAPI, Prometheus, eBPF, TLS — *24-hr Challenge*

- Designed distributed system architecture for agent coordination with fault-tolerant pipelines and secure communication.
- Developed async communication layers and CLI tooling to optimize inter-agent workflows with real-time observability.

AI-Powered Object Detection for Smart Cities — Oct 2024 – Dec 2024
YOLOv8, OpenCV, Pandas, Matplotlib — *University Project*

- Built real-time computer vision pipeline for traffic and pedestrian detection with RTOS integration for edge deployment.
- Generated data-driven insights and visualizations for urban planning applications.

Education

VIT Bhopal University — Sehore, India
Integrated M.Tech in Artificial Intelligence — Expected Graduation: 2028 — *Sep 2023 – Present*